

Stochastic Mathematics for Finance

Week 1 Assignment

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Question 1: Portfolio Loss Dependence

Given:

$$P(L_A) = 0.10$$

$$P(L_B) = 0.20$$

$$P(L_A|L_B) = 0.35$$

(a) Compute $P(L_A \cap L_B)$ and $P(L_A \cup L_B)$

Using:

$$P(L_A \cap L_B) = P(L_A|L_B)P(L_B)$$

$$= 0.35(0.20)$$

$$= 0.07$$

Now,

$$P(L_A \cup L_B) = P(L_A) + P(L_B) - P(L_A \cap L_B)$$

$$= 0.10 + 0.20 - 0.07$$

$$= 0.23$$

(b) Independence Check

If independent:

$$P(L_A \cap L_B) = P(L_A)P(L_B)$$

$$= 0.10(0.20) = 0.02$$

Since:

$$0.07 \neq 0.02$$

events are not independent.

(c) Compute $E[N]$ and $Var(N)$

Let:

$$N = I_A + I_B$$

Expectation:

$$E[N] = E[I_A] + E[I_B]$$

$$= 0.10 + 0.20$$

$$= 0.30$$

Variance:

$$Var(I_A) = 0.10(0.90) = 0.09$$

$$Var(I_B) = 0.20(0.80) = 0.16$$

Covariance:

$$Cov(I_A, I_B) = 0.07 - (0.10)(0.20)$$

$$= 0.05$$

Therefore:

$$Var(N) = 0.09 + 0.16 + 2(0.05)$$

$$= 0.35$$

(d) Naive Independence

Joint probability under independence:

$$0.10(0.20) = 0.02$$

Factor:

$$\frac{0.07}{0.02} = 3.5$$

Thus the independence assumption underestimates risk by factor 3.5.

(e) Importance of Dependence

Dependence between losses increases portfolio risk because multiple assets may lose value together.

(f-g) Simulation Results

Insert plot below:

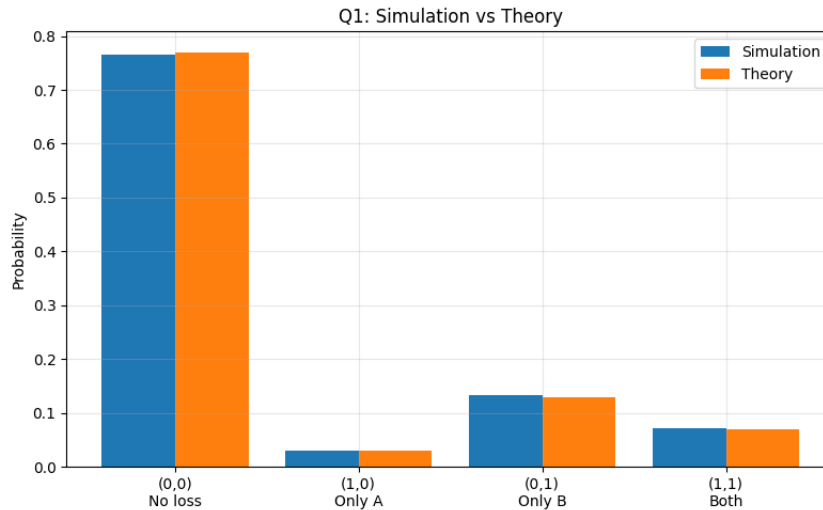


Figure 1: Simulation vs theoretical values

Quantity	Theory	Simulation
$P(L_A \cap L_B)$	0.07	0.0706
$P(L_A \cup L_B)$	0.23	0.2335
$E[N]$	0.30	0.3041
$Var(N)$	0.35	0.3528

Question 2: Credit Rating Migration

States:

$$S = \{0, 1, 2, 3\}$$

where:

0 = Investment Grade

1 = Speculative Grade

2 = Distressed

3 = Default

Transition matrix:

$$P = \begin{pmatrix} 0.60 & 0.25 & 0.05 & 0.10 \\ 0 & 0.50 & 0.50 & 0 \\ 0 & 0.50 & 0.50 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(a) Directed graph

- State 0 connects to states 0, 1, 2, 3
- States 1 and 2 communicate with each other
- State 3 is absorbing

(b) Communicating classes

- $\{0\}$: open class
- $\{1, 2\}$: closed communicating class
- $\{3\}$: closed absorbing class

(c) Recurrent and transient states

- States 1, 2, 3 are recurrent
- State 0 is transient

(d) Irreducibility and aperiodicity

The chain is not irreducible since multiple communicating classes exist.

Class $\{1, 2\}$ is aperiodic because self-loops exist.

Class $\{3\}$ is also aperiodic.

(e) Stationary distributions

For class $\{3\}$:

$$\pi_3 = (0, 0, 0, 1)$$

For class $\{1, 2\}$:

$$P_{12} = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$$

Solving:

$$(\pi_1, \pi_2)P_{12} = (\pi_1, \pi_2)$$

with:

$$\pi_1 + \pi_2 = 1$$

gives:

$$\pi_{12} = (0, 0.5, 0.5, 0)$$

(f) Full stationary family

$$\pi(\alpha) = \alpha(0, 0.5, 0.5, 0) + (1 - \alpha)(0, 0, 0, 1)$$

where:

$$0 \leq \alpha \leq 1$$

(g-h) Simulation results

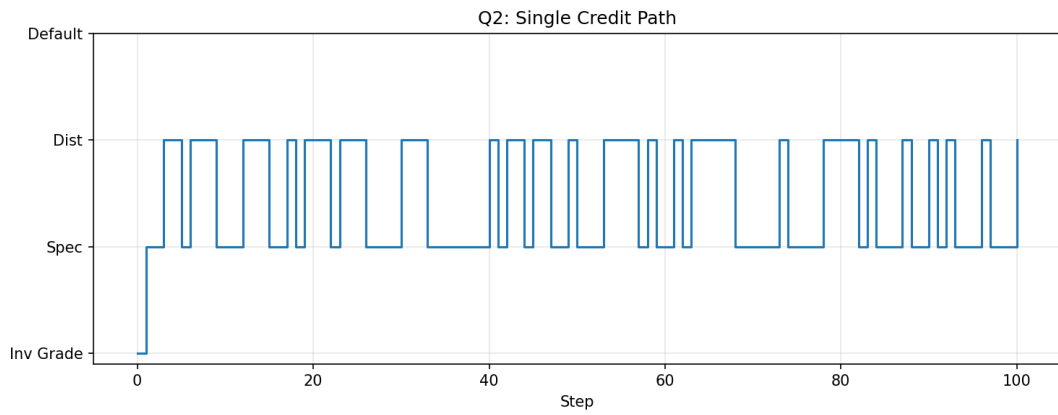


Figure 2: Single simulated credit migration path

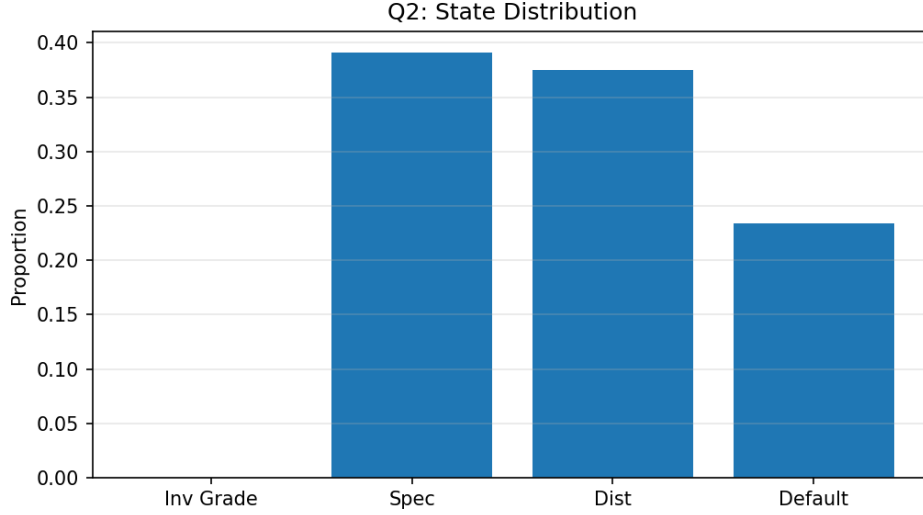


Figure 3: Distribution of states at time $t = 100$

State	Proportion
Investment Grade	0.000
Speculative	0.390
Distressed	0.375
Default	0.235

Table 1: Final state distribution at $t = 100$

(i) Interpretation

- Class $\{1, 2\}$ represents persistent movement between speculative and distressed states.
- Class $\{3\}$ is absorbing since default cannot be exited.
- State 0 is transient because the chain eventually leaves it.

Question 3: Inventory Random Walks

(a) Financial interpretation

- Recurrent origin: inventory returns to the neutral position repeatedly.
- Transient origin: inventory drifts away from neutrality.

(b) Recurrence criterion

A state i is recurrent if:

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$$

and transient if:

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty$$

(c) Recurrence and transience

For a simple symmetric random walk:

$$p_{00}^{(2n)} \sim \frac{C_d}{n^{d/2}}$$

For $d = 1$:

$$\sum \frac{1}{\sqrt{n}} = \infty$$

Hence recurrent.

For $d = 2$:

$$\sum \frac{1}{n} = \infty$$

Hence recurrent.

For $d = 3$:

$$\sum \frac{1}{n^{3/2}} < \infty$$

Hence transient.

(d-f) Simulation results

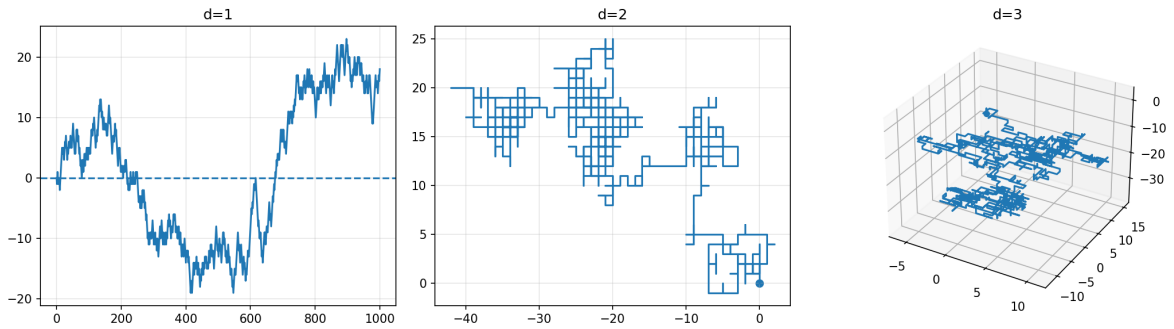


Figure 4: Sample random walk paths for $d = 1, 2, 3$ (1000 steps)

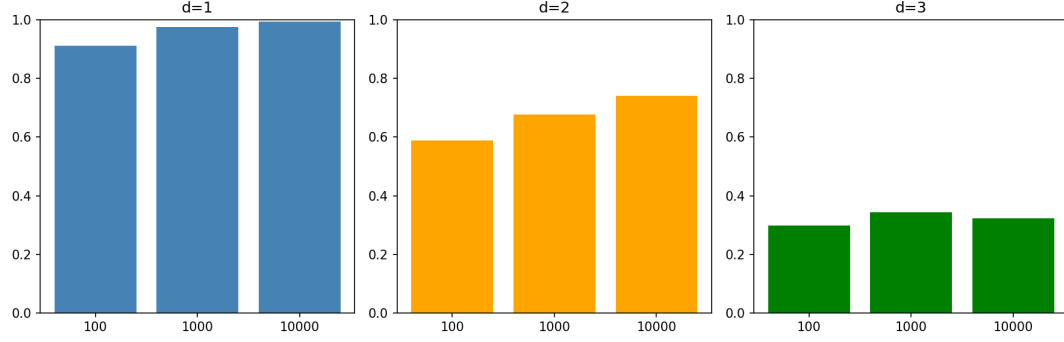


Figure 5: Return probability vs path length

Path length	$d = 1$	$d = 2$	$d = 3$
100	0.912	0.588	0.299
1000	0.975	0.676	0.344
10000	0.994	0.740	0.323

Table 2: Return proportions

(g) Interpretation

Simulation suggests recurrence or transience but cannot prove it since only finite paths are observed.

Question 4: Market Regime Switching

States:

0 = Bull 1 = Neutral 2 = Bear 3 = Crisis

Returns:

$$r = (0.012, 0.004, -0.006, -0.025)$$

(a) Irreducibility and aperiodicity

The chain is irreducible since every state can reach every other state.

Since all diagonal entries are positive, the chain is aperiodic.

(b-c) Stationary distribution

$$\pi = (0.2524, 0.3442, 0.2390, 0.1644)$$

π_i gives the long-run fraction of time spent in regime i .

(d-e) Long-run return

$$\sum_i \pi_i r_i = -0.001138$$

Long-run unfavourable fraction:

$$\pi_2 + \pi_3 = 0.4034$$

(f-i) Simulation results

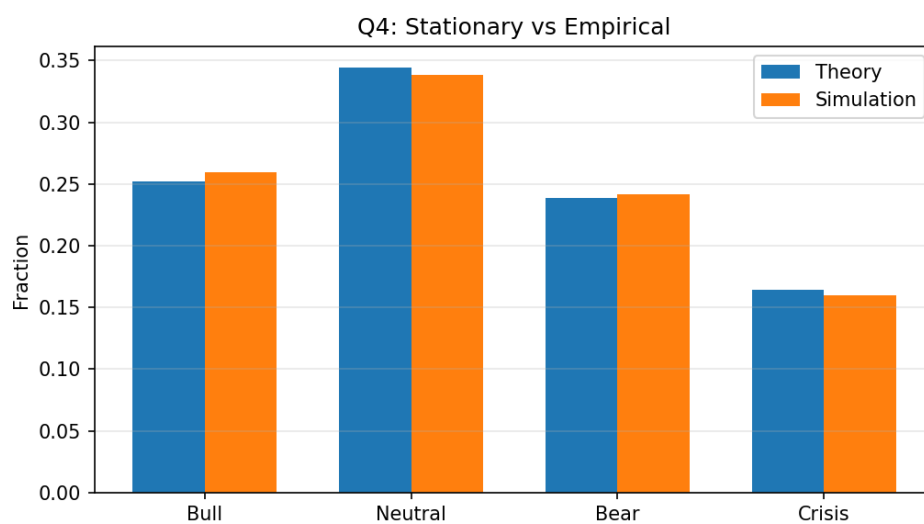


Figure 6: Stationary vs empirical frequencies

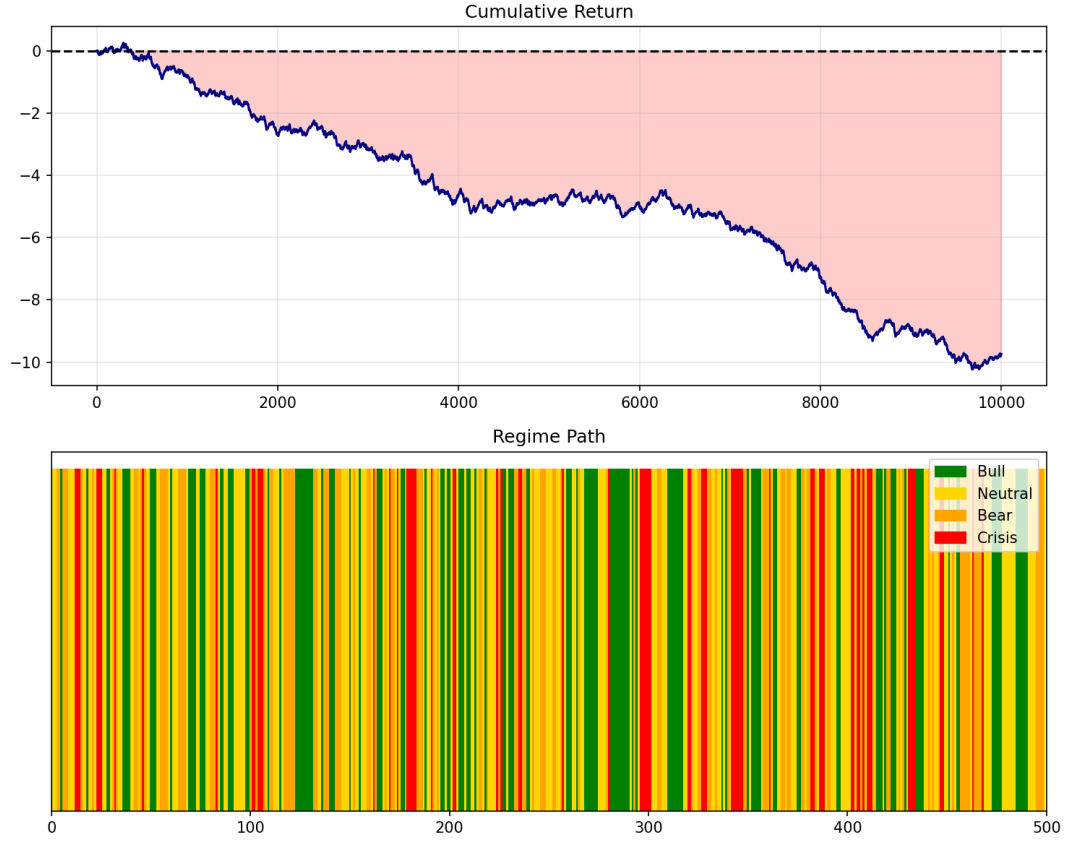


Figure 7: Cumulative return

Regime	Stationary	Empirical
Bull	0.2524	0.2598
Neutral	0.3442	0.3386
Bear	0.2390	0.2418
Crisis	0.1644	0.1598